

Daniyar Tolegenov

Senior ML Engineer • NLP

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SUMMARY

NLP-focused ML engineer with 5 years of experience building and fine-tuning transformer models for production. Specialized in multilingual systems (Russian, Kazakh, English) for text classification, NER, and generation. Hands-on experience with PEFT fine-tuning (LoRA/QLoRA), model deployment via ONNX/Triton, and dataset curation.

EXPERIENCE

Senior NLP Engineer | alem.ai (Almaty)

April 2023 — Present

- Fine-tuned multilingual BERT and XLM-RoBERTa for sentiment analysis on Kazakh-language social media data: accuracy 0.89 on held-out test (3K labeled examples).
- Trained custom NER model for extracting entities from legal documents (Kazakh + Russian): organizations, persons, monetary values, dates. F1=0.85 micro-avg.
- Built dataset curation pipeline using Label Studio + active learning: reduced labeling cost by 60% vs random sampling.
- Productionized models via ONNX export + Triton Inference Server. p95 latency 35ms for BERT-base.
- Fine-tuned Qwen2-7B with QLoRA on internal corpus for instruction-following: deployed via vLLM, throughput 80 req/s on a single A100.
- Implemented model monitoring: token distribution drift, output quality metrics, automated alerting.

NLP Engineer | Mail.ru Group (remote)

Aug 2020 — March 2023

- Worked on the My.Tracker analytics platform: text classification for user feedback (10M+ tickets per month).
- Distilled BERT-large into a smaller model (8x speedup) with <1% accuracy loss using HuggingFace distillation pipeline.
- Built embedding-based retrieval system using sentence-transformers + FAISS for support agent suggestions.
- Wrote production code for tokenization, preprocessing, postprocessing of Russian-language input.

ML Engineer (Intern → Junior) | Tinkoff (remote)

June 2019 — July 2020

- Worked on intent classification for the chatbot serving banking customers.
- Data pipelines on Spark, model serving on Kubernetes.
- Wrote evaluation scripts and dashboards in Streamlit.

TECHNICAL SKILLS

Core: Python (5 years), PyTorch (advanced), TensorFlow (intermediate)
NLP: HuggingFace Transformers, tokenizers, datasets, accelerate, sentence-transformers
Fine-tuning: PEFT (LoRA, QLoRA, IA3), TRL (SFT, RLHF basics), bitsandbytes 4-bit/8-bit
Serving: vLLM, Triton Inference Server, ONNX Runtime, TorchServe basics
Architectures: BERT family, RoBERTa, T5, mT5, GPT, LLaMA, Qwen, Mistral
Multilingual: Russian, Kazakh, English production experience
Data: Label Studio, active learning, data augmentation (back-translation, NLPAug)
Infrastructure: Docker, Kubernetes, MLflow, DVC, Git, Linux

EDUCATION

Higher School of Economics (HSE)

MSc Data Science (NLP track) • 2018 — 2020

Nazarbayev University

BSc Computer Science • 2014 — 2018

PUBLICATIONS

- Tolegenov, D. et al. "Cross-lingual Transfer for Low-resource NER in Kazakh." Workshop on NLP for Turkic Languages, 2023.
- Open-source: maintainer of kaz-nlp-tools (Python library for Kazakh language preprocessing, 200+ stars on GitHub).

LANGUAGES

Kazakh — native • Russian — native • English — C1 (active conference speaker)